

Module 14) Python – Collections, functions and Modules

2. List Operations

- **Common list operations: concatenation, repetition, membership**

Concatenation

Definition:

The process of combining two or more lists to create a new list.

Key Characteristics:

- Uses the + operator
- Does not modify the original lists
- Creates a new list containing all elements from both lists in order
- Time complexity: $O(n)$ where n is the total number of elements

Repetition

Definition:

Creating a new list by repeating the elements of an existing list multiple times.

Key Characteristics:

- Uses the * operator with an integer
- The original list remains unchanged
- The integer determines how many times the list is repeated
- Time complexity: $O(n*k)$ where n is list length and k is repetition count

Membership Testing

Definition:

Checking whether an element exists in a list.

Key Characteristics:

- Uses in and not in operators
- Returns a boolean value (True/False)
- Performs linear search (checks elements one by one)
- Time complexity: $O(n)$ in worst case
- Case-sensitive for strings
- Uses equality comparison (==) for matching

- **Understanding list methods like `append()`, `insert()`, `remove()`, `pop()`**

`append()`

Purpose:

Adds a single element to the end of the list.

Characteristics:

- Modifies the original list
- Accepts exactly one argument
- Time complexity: $O(1)$ (amortized)
- No return value (returns None)

`list.append(element) → None` # list now has element at end

insert()

Purpose:

Inserts an element at a specified position.

Characteristics:

- Takes two arguments: index and element
- Shifts subsequent elements to the right
- Time complexity: $O(n)$ in worst case
- Accepts negative indices (counts from end)
- No return value (returns None)

remove()

Purpose:

Removes the first occurrence of a specified value.

Characteristics:

- Takes one argument: value to remove
- Raises ValueError if element not found
- Time complexity: $O(n)$
- Performs linear search for the value
- No return value (returns None)

pop()

Purpose:

Removes and returns an element at a given index.

Characteristics:

- Optional index argument (default: -1)
- Returns the removed element
- Raises IndexError for invalid indices

- Time complexity: $O(1)$ for end, $O(n)$ for other positions
- Without arguments, removes last element